Finding Zeros with the Rational Root Theorem
Answer Key
Algebra 2

Quick Answers

1. 0
2. -1, 2, 3
3. $(x-2)\,(x+1)\,(x+3)$
4. 0
5. $\frac{1}{2}$, 2, -2
6. 2, $1+\sqrt{3}$, $1-\sqrt{3}$
7. $(x-2)\,(x+2)\,(3x-1)$
8. -3, $2+\sqrt{5}$, $2-\sqrt{5}$
9. -2, -1, 2, 3
10. 1, $2i$, $-2i$
11. $(x-2)\,(2x-1)\,(2x+1)$
12. —

Curriculum

Level: Algebra 2
HSA-APR.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 12 tagged checks.

1. $f(x) = x^3 - 4x^2 + x + 6$**; test** $x = 2$**.**

$$f(2) = 8 - 16 + 2 + 6 = 0$$

A remainder of zero is exactly what "$x = 2$ is a zero" means, so $(x-2)$ is a factor of f. $\boxed{f(2) = 0}$

2. All zeros of $f(x) = x^3 - 4x^2 + x + 6$**.** Divide out the factor found in problem 1. Synthetic division by $x - 2$ on the coefficients $1,\ -4,\ 1,\ 6$ gives the quotient coefficients $1,\ -2,\ -3$ with remainder 0:

$$f(x) = (x-2)(x^2 - 2x - 3) = (x-2)(x-3)(x+1)$$

Set each factor to zero. $\boxed{x = -1,\ x = 2,\ x = 3}$
Check: the three zeros multiply to $-1 \cdot 2 \cdot 3 = -6$, and for a monic cubic that product must equal $-(\text{constant term}) = -6$. ✓

3. Factor $g(x) = x^3 + 2x^2 - 5x - 6$**.** Constant -6, leading coefficient 1, so the candidates are $\pm1, \pm2, \pm3, \pm6$. $g(1) = -8 \neq 0$; $g(-1) = -1 + 2 + 5 - 6 = 0$, so $(x+1)$ is a factor. Dividing gives $x^2 + x - 6$, which factors by inspection.

$$g(x) = (x+1)(x^2 + x - 6)$$

$\boxed{(x+1)(x+3)(x-2)}$

4. $h(x) = 2x^3 - x^2 - 8x + 4$**; test** $x = \frac{1}{2}$**.** With leading coefficient 2, the theorem allows $q \in \{1, 2\}$, so halves join the candidate list.

$$h\left(\tfrac{1}{2}\right) = 2 \cdot \tfrac{1}{8} - \tfrac{1}{4} - 4 + 4 = \tfrac{1}{4} - \tfrac{1}{4} = 0$$

So $x = \frac{1}{2}$ is a zero and $(2x - 1)$ is a factor. $\boxed{h(\tfrac{1}{2}) = 0}$

5. All zeros of $h(x) = 2x^3 - x^2 - 8x + 4$**.** Divide out the factor from problem 4:

$$h(x) = (2x - 1)(x^2 - 4) = (2x - 1)(x - 2)(x + 2)$$

The difference of squares finishes it with no further division needed. $\boxed{x = \tfrac{1}{2},\ x = 2,\ x = -2}$

Note the form of the factor: the zero $\frac{1}{2}$ corresponds to the factor $(2x - 1)$, not $(x - \frac{1}{2})$, when you keep integer coefficients.

6. All zeros of $p(x) = x^3 - 4x^2 + 2x + 4$**.** Candidates $\pm 1, \pm 2, \pm 4$. $p(1) = 3$, $p(-1) = -3$, $p(2) = 8 - 16 + 4 + 4 = 0$. Divide by $x - 2$: the quotient is $x^2 - 2x - 2$, which has no rational zeros, so use the quadratic formula:

$$x = \frac{2 \pm \sqrt{4 + 8}}{2} = \frac{2 \pm 2\sqrt{3}}{2} = 1 \pm \sqrt{3}$$

$\boxed{x = 2,\ x = 1 + \sqrt{3},\ x = 1 - \sqrt{3}}$

Why the theorem could not find the last two: $1 \pm \sqrt{3}$ is irrational, and the theorem only ever produces rational candidates.

7. Factor $q(x) = 3x^3 - x^2 - 12x + 4$**.** Candidates are $\pm 1, \pm 2, \pm 4, \pm \frac{1}{3}, \pm \frac{2}{3}, \pm \frac{4}{3}$. Testing $x = \frac{1}{3}$: $3 \cdot \frac{1}{27} - \frac{1}{9} - 4 + 4 = 0$, so $(3x - 1)$ is a factor. Division leaves $x^2 - 4$:

$$q(x) = (3x - 1)(x^2 - 4)$$

$\boxed{(3x - 1)(x - 2)(x + 2)}$

Shortcut worth naming: grouping also works here — $x^2(3x - 1) - 4(3x - 1)$ — and grouping is always worth a look when the terms pair up like this.

8. Solve $x^3 - x^2 - 13x - 3 = 0$**.** Candidates $\pm 1, \pm 3$. Substituting: -16, 9, -60, and at $x = -3$, $-27 - 9 + 39 - 3 = 0$. So $(x + 3)$ divides the cubic, and synthetic division gives the quotient $x^2 - 4x - 1$:

$$x = \frac{4 \pm \sqrt{16 + 4}}{2} = \frac{4 \pm 2\sqrt{5}}{2} = 2 \pm \sqrt{5}$$

$\boxed{x = -3,\ x = 2 + \sqrt{5},\ x = 2 - \sqrt{5}}$

9. All zeros of $r(x) = x^4 - 2x^3 - 7x^2 + 8x + 12$. Candidates $\pm1, \pm2, \pm3, \pm4, \pm6, \pm12$. $r(-1) = 1 + 2 - 7 - 8 + 12 = 0$, so divide by $x + 1$:

$$r(x) = (x + 1)(x^3 - 3x^2 - 4x + 12)$$

Run the routine again on the cubic: its candidates are the divisors of 12, and $x = 2$ gives $8 - 12 - 8 + 12 = 0$. Dividing again leaves $x^2 - x - 6$, which factors as $(x - 3)(x + 2)$:

$$r(x) = (x + 1)(x - 2)(x - 3)(x + 2)$$

$\boxed{x = -2,\ x = -1,\ x = 2,\ x = 3}$

10. All zeros of $s(x) = x^3 - x^2 + 4x - 4$, including non-real ones. Candidates $\pm1, \pm2, \pm4$. $s(1) = 1 - 1 + 4 - 4 = 0$, so $(x - 1)$ is a factor, and grouping confirms it: $x^2(x-1)+4(x-1) = (x-1)(x^2+4)$. Then $x^2 = -4$ gives $x = \pm 2i$. $\boxed{x = 1,\ x = 2i,\ x = -2i}$
Answer to the second question: the theorem supplied exactly one zero, the rational one. The other two are non-real, and non-real zeros are never on a candidate list. The count still works out: degree 3, three zeros.

11. Factor $t(x) = 4x^3 - 8x^2 - x + 2$. Candidates $\pm1, \pm2, \pm\frac{1}{2}, \pm\frac{1}{4}$. $t(2) = 32 - 32 - 2 + 2 = 0$, so $(x - 2)$ is a factor and division leaves $4x^2 - 1$, a difference of squares:

$$t(x) = (x - 2)(4x^2 - 1)$$

$\boxed{(x - 2)(2x - 1)(2x + 1)}$

12. Manual review — expected reasoning for $u(x) = x^3 - 3x - 1$. Two points must appear.

1. The theorem is a statement about *rational* zeros only. Exhausting the candidate list proves u has no rational zero; it says nothing about irrational or non-real zeros. (In fact all three zeros of this cubic are real and irrational.)

2. Evidence of a real zero: $u(-1) = 1 > 0$ and $u(0) = -1 < 0$. A polynomial is continuous, so a sign change on $[-1, 0]$ forces a zero in between — the Intermediate Value Theorem. A graph or a table showing the same sign change is equally acceptable.

Accept any answer making both points. Do not accept "the theorem must have missed one" with no argument. A student who adds that an odd-degree polynomial always has at least one real zero has given a second valid argument.